NOMENCLATURE OF ORGANIC COMPOUNDS

PYQ • 2021 - 2026

1. 2026 **1 MARK**

Match the following :

A : (X) –CHO (Y) –O–R (Z) –OH

B : (i) Ether (ii) Aldehyde (iii) Keto (iv) Hydroxyl

Choose the correct answer from the options given below.

(a) (X) (ii) (Y) (iii) (Z) (iv)

(b) (X) (iv) (Y) (i) (Z) (ii)

(c) (X) (ii) (Y) (i) (Z) (iv)

(d) (X) (ii) (Y) (iv) (Z) (i)

2. 2026 **2 MARKS**

The molecular formulae of the first three members of a homologous series are given.

C_2H_2, C_3H_4, C_4H_6

(a) Write the molecular formula of the fifth member of this series.

(b) Write the structural formula of a position isomer of this compound.

3. 2026 **3 MARKS**

Some hints about an organic compound are given.

(i) There are five carbon atoms in one molecule.

(ii) The position of –CO– group is two.

(a) Write the structural formula of this organic compound.

(b) Write the structural formula and IUPAC name of a functional isomer of this compound.

4. 2025 **4 MARKS**

Match the following suitably.

A : (i) $CH_3–CH(CH_3)–CH(CH_3)–CH_3$ (ii) $CH_3–CH=CH–CH_3$

(iii) $CH_3–CH(OH)–CH_3$ (iv) $CH_3–CH_2–CH_2–CH_2–COOH$

B : C_3H_8O, C_6H_{14}, $C_5H_{10}O_2$, C_4H_8

C : Pentanoic acid, Propan-2-ol, But-2-ene, 2,3-Dimethylbutane

5. 2025 **2 MARKS**

The chemical formula of a compound is C_2H_6O.

(a) Write the structural formula of any one of the functional isomers.

(b) Write the IUPAC name of this functional isomer.

6. 2025 **2 MARKS**

Classify the following compounds into alkanes, alkenes and alkynes.

C_2H_4, C_5H_{12}, C_3H_8, C_7H_{12}

7. 2024 **4 MARKS**

The structural formula of an organic compound is given :

$CH_3–CH_2–O–CH_3$

(a) The compounds having –O–R as the functional group are called _____ .

(b) Write the molecular formula of this compound.

(c) Write the IUPAC name of this compound.

(d) Write the structural formula of the functional isomer of this compound.

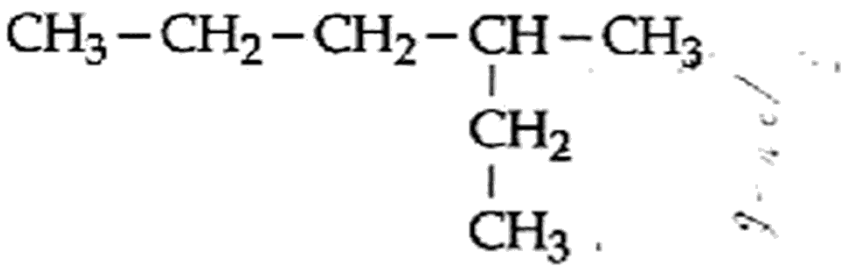
8. 2024 **3 MARKS**

The structural formula of a hydrocarbon is given below.

(a) How many carbon atoms are there in the main chain ?

(b) Name the branch present in the compound.

(c) Write the IUPAC name of this compound.



9. 2024 **2 MARKS**

The molecular formula of a hydrocarbon is C_4H_8.

(a) To which category does this hydrocarbon belong ?

[Alkane, Alkene, Alkyne]

(b) Draw the structure of an alicyclic hydrocarbon with the same molecular formula.

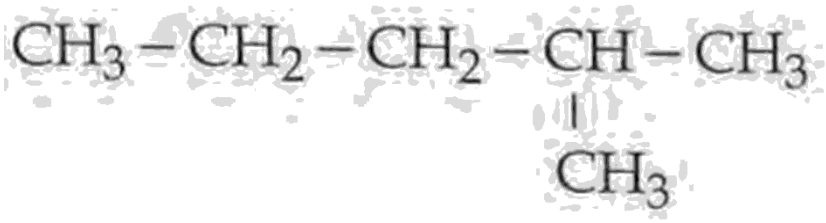
10. 2022 **4 MARKS**

(a) How many carbon atoms are there in the longest chain of this hydrocarbon ?

(b) Give the name of the branch.

(c) What is the position number of the branch ?

(d) Write the IUPAC name of the compound.



11. 2022 **3 MARKS**

(i) $CH_3–CH_2–CH_2–CH_3$

(ii) $CH_3–CH_2–CH_2–CH_2–OH$

(iii) $CH_3–CH_2–O–CH_2–CH_3$

(iv) $CH_3–CH_2–CH_3$

(a) Identify the isomer pair in the given compounds.

(b) Name the isomerism.

(c) How many isomers are possible for compound (i) ?

12. 2022 **1 MARK**

Identify the compound which contains a carbon-carbon triple bond.

(C_5H_{12}, C_2H_2, C_3H_6, CH_4)

13. 2021 **4 MARKS**

The structural formulae of two organic compounds are given.

(i) $CH_3–O–CH_2–CH_3$ (ii) $CH_3–CH_2–CH_2–OH$

(a) Write the IUPAC name of the first compound.

(b) These two organic compounds are isomers. Why ?

(c) Which type of isomerism is exhibited by them ?

(d) Write the structural formula of the position isomer of the second compound.

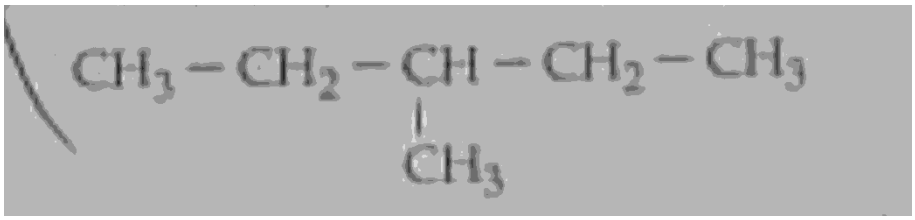
14. 2021 **3 MARKS**

The structural formula of a hydrocarbon is given.

(a) How many carbon atoms are there in the main chain ?

(b) What is the position number of the branch ?

(c) Write the IUPAC name of this compound.

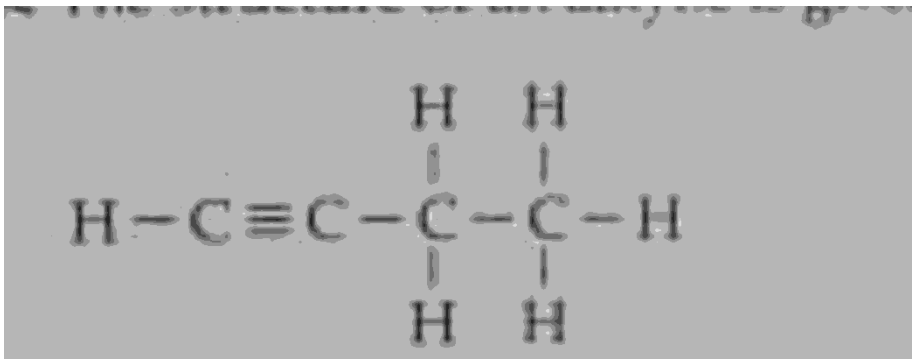


15. 2021 **2 MARKS**

The structure of an alkyne is given.

(a) What is the molecular formula of this compound ?

(b) Write the IUPAC name.



16. 2021 **2 MARKS**

CH_4, C_2H_6, C_3H_8 and C_4H_{10} are the members of a homologous series.

(a) What is the molecular formula of the 6th member of this series ?

(b) Write the general formula of this homologous series.

17. 2021 **1 MARK**

Identify the alkene among the given hydrocarbons.

(C_2H_6, C_2H_4, C_2H_2, CH_4)